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RESEARCH ARTICLE - ELECTRICAL ENGINEERING

Improvement of Handover Performance Based on Bio-Inspired Approach with Received Signal Strength and Mean Opinion Score

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Abstract In mobile communication environments, handover is a very important process to maintain mobile host's connections to the network. Although a lot of researchers have tried to improve the handover performance, little deployment to satisfy the users can be seen in the real network due to the modification costs. In the current age, the existence of network with different radio access technologies has created a heterogeneous environment. Moreover, the mobile hosts available to users are becoming more sophisticated, which opens the opportunity for end-point approaches to alleviate the deployment issues. Furthermore, real-time applications such as VoIP and video conference are delay and quality sensitive, but not much research has considered qualitative requirement. Hence, a suitable technique is needed to alleviate the deployment issues as well as to cope with the stringent requirements of real-time applications. This paper presents a new end-point centric handover approach based on the ant colony probabilistic equation that utilizes the received signal strength and the mean opinion score to trigger the handover. Some simulations were conducted using OMNeT++ to measure the effectiveness of the proposed approach comparing with an existing method, the Endpoint Centric Handover (ECHO). The results show that the proposed approach has better performance compared to the ECHO.

Keywords Handover · End-point centric · Mobile network · Mobility management · Host identity protocol · Bio-inspired · Ant colony probabilistic equation

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1 Introduction

When considering a mobile network scenario, the handover is a very important process in maintaining the mobile host (MH) connection to the network. Especially in the current age, a lot of users own at least one or more mobile devices such as smart phones and mobile tablets which are on par computational capabilities with laptop or desktop computers. This is because of the usefulness, ubiquity and comparably affordable price; consequentially inciting the development of wireless access technologies. Hence, the rapid evolution of radio access technology (RAT) has been done including some new systems such as Worldwide Interoperability for Microwave Access (WiMAX) [1] and Long-Term Evolution (LTE) [2] which have been deployed nearly worldwide, adding more spice to the pot of the currently available wireless networks such as legacy cellular networks and the IEEE802.11-based wireless local area networks (WLAN). The integration of these wireless access networks has structured a heterogeneous environment. Since most of the mobile devices are capable of connecting to at least two different RATs (i.e. WLAN, cellular, WiMAX, LTE and so forth), there is a growing demand for ubiquitous access to the Internet via various different RATs.

Furthermore, with the high capabilities of the existing mobile equipment, the users gain access to a slew of applications that requires the Internet connection. The connection

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requirements for these applications vary; simple applications such as messaging, e-mail and browsing are usually delay tolerant and does not require very high computational needs. On the contrary, real-time applications, in particular, Voice over IP (VoIP) [3] and Video Conference, have very stringent requirements due to its delay sensitiveness and its qualitative requirements, not only for the Quality of Service (QoS) of the network, but also for the Quality of Experience (QoE) of the users. Hence, to efficiently utilize the existing heterogeneous networks while maintaining the QoS and QoE, a suitable handover approach is required; in this case, a vertical handover (VHO) approach, which is necessary for multiple different RAT must be well-considered.

The vertical handover process is basically divided into three phases; the Handover Information Gathering Phase (HIGP), the Handover Decision Phase (HDP), and finally the Handover Execution Phase (HEP). A lot of studies on the HDP have been reported which decides the *where* and *when* to trigger the handover. The surveys reported in [4] and [5] have shown many parameters such as received signal strength (RSS) and other parameters including network contexts, user preferences, mobile terminal's mobility and other VHO contexts. The most important information for real-time applications like VoIP is the expected level of users' QoE. One of the approaches that utilize the QoE metric has been proposed by [6]. Some input parameters used to determine the best network and to trigger the handover are the bandwidth, network speed and received signal strength. However, these parameters do not directly portray the QoE that a user can expect. In real-time applications such as VoIP, one metric that directly shows the QoE level is the ITU-T E-Model's mean opinion score (MOS). This information is very useful to determine the best network, from a user's point of view, that can maintain the requirements of VoIP as shown in [7]. Nevertheless, not many works have considered MOS in HDP.

Furthermore, from the literature, the HEP can be divided into two parts which are the Handover Management (HM) and the Mobility Management (MM). There are several types of HMs such as Mobile-controlled handoff (MCHO) where the MH is the one that decides and triggers the handover, Network-controlled handoff (NCHO) where the handover is totally controlled by the network equipment such as the base station and mobile-assisted handoff (MAHO) where the handover is initiated and decided by the network entity, but with help and information from the MH [8]. Meanwhile, the MM is mostly depending on mobility management protocols. The popular protocols are Mobile IPv4 (MIPv4) and its variants such as Mobile IPv6 (MIPv6), Hierarchical MIP (HMIP), Fast MIP (FMIP), and Fast Hierarchical MIP (FHMIP). These protocols have been researched extensively and have shown good performance in research. Nevertheless, even though MIP enabled equipment have been available for some time, not much deployment of these technologies can be seen in the real network currently [9]. This is because huge modifications have to be done to the existing networks that will incur a very massive cost [9].

Consequently, there is a growing interest in the research communities in studying end-point centric approaches to avoid or minimize the alterations in the network entities [10]. This approach is possible because of the rapid advancement of mobile hosts; with the capability of the existing and future mobile devices, the complexity of NCHO can be pushed to the end-points (mobile devices in this case) in order to minimize or totally avoid network modifications. Several existing methods currently available are the Host base autonomous Mobile Address Translation (HaMAT) [11], the Seamless IP diversity-based Generalized Mobility Architecture (SIGMA) [12–14] and the Endpoint Centric H and over (ECHO) [7,15,16]. These methods are very similar in terms of the implementation albeit using different protocols. HaMAT uses Mobile Address Translation (MAT) based on the network layer, while both SIGMA and ECHO use Stream Control Transmission Protocol (SCTP) (these are the MM components). The similarity is that all of them use multihoming (where the mobile node will be able to connect to two or more networks at the same time via multiple antennas of different RATs), the availability of multiple network connections (the term "path" will also be used in this manuscript) between the MH and its Corresponding Host (CH) before any handover is needed, and all of them are obviously end-point centric. However, in [11], there are no discussions on how the HDP is done because the study focused on implementing the HaMAT; SIGMA uses only RSS value to trigger the handover, while ECHO, which is based on SIGMA, enhanced the triggering with an additional QoE factor, but implemented only static thresholds for the QoE and RSS value which restrict the method. In our previous work [17,18], which is the foundation of the work in this manuscript, we have studied the effectiveness of using Host Identity Protocol (HIP) compared to MIP-based handover approaches. In that study, the handover is simply triggered using the RSS and the results have shown that HIP is capable to maintain low packet loss even when the MH is moving at high velocity. However, no consideration on the QoE has been included.

Based on our previous work, this paper proposes a new approach to improve the disadvantages of the existing end-point centric handover. The proposed approach utilizes the RSS value and the QoE information which is obtained by using the MOS [19], similar to [15] but with different algorithm for HDP and uses the HIP. In the previous work, static values were used as threshold to determine the handover triggering. On the other hand, in the proposed method, the concept of Ant Colony [20] that implements the probabilistic equation to combine the contexts for the handover triggering is used. Furthermore, the use of weights assigned to the parameters and the probability comparison can allevi-

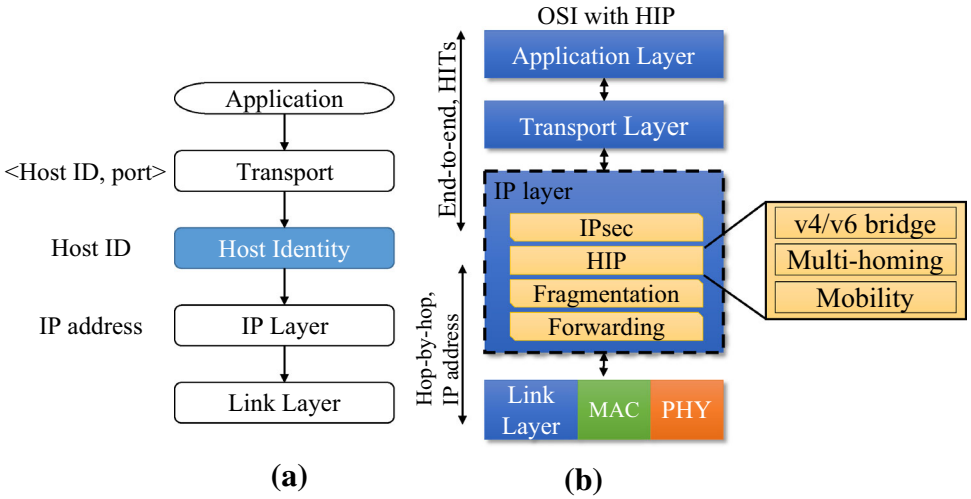


Fig. 1 The host identity protocol: **a** the idea of HIP, host identity is introduced as the identifier. **b** Detailed layering of HIP, provides multi-homing and mobility

ate the restriction of using static threshold in coping with the dynamic changes in a wireless network. The effectiveness of the proposed method is validated by comparing the simulation results of ECHO with the proposed method using OMNeT++ [21].

The rest of this paper is organized as follows: Sect. 2 briefly describes the Ant Colony concept, HIP and MOS, followed by the proposed handover approach in Sect. 3. Section 4 presents the simulation results and the comparison between ECHO and the proposed method. Finally, the paper is concluded in Sect. 5.

2 Methodology

This section will discuss the main components used in the proposed method. Due to the use of end-point centric approach, the usable contexts are constricted to the information attainable by the MH. The proposed approach exploits the RSS of the MH combined with the MOS level information using the Ant Colony Probabilistic Equation (ACPE) to trigger the handover. As discussed in the previous section, the HEP is divided into HM and MM; in the proposed method, MCHO scheme is opted and HIP is used for the MM. In the following subsections, the overview of HIP, MOS and how it is obtained, as well as how ACPE is used to trigger the handover are discussed. An outline of ECHO [7], which is a close research to this work, will also be included in order to easily understand and discuss the comparison.

2.1 Host Identity Protocol

The Host Identity Protocol is a quite new protocol which was proposed by Moskowitz et al. [22]. In the current Internet system, the IP address not only works as the location address (locators), but also works as the identity or the ID of the end-point hosts (identifiers). As illustrated in Fig. 1a, HIP introduces a new Name Space of Host Identifiers (HI) so that sockets of applications or transport are bounded to these HIs instead of the IP addresses. The introduction of HI separates the function of the IP address, where the HI will work as the identifier, while the IP address will only work as the locator as shown by Fig. 1b. This is useful in mobile communications because the identifier does not change even though the MH changes its IP address when moving into a new network; making the layers below HIP to become transparent to the upper layers (transport, applications and etc.). Furthermore, HIP enables the MH to use multi-homing. That means, the MH will be able to connect to two or more networks at the same time depending on the number of Network Interface Cards (NIC). Besides that, HIP supports mobility management via the rendezvous server and the UPDATE message [23].

The update sequence used by HIP is as shown in Fig. 2. Referring to this figure, when the MH moves into an overlapped area between AP1 (access point 1) and AP2, the second NIC of the MH will associate with the new network. Subsequently, when the IP address for the new network is obtained, the first UPDATE message will be sent to the CH to give the additional IP address. When the CH receives this message, it can quickly start communicating via the new network location but with some restrictions according to [24]. The CH will send back an UPDATE message to acknowledge the first message including an echo request so that when the MH receives this message it will send another message to the CH to confirm the connection to the new network. When the MH receives the UPDATE from the CH, it can start sending IP packets via AP2. Take note that all these processes are communicated and processed on the HIP layer. If the MOS is measured at the application layer, it is difficult to determine the different packet transfer path available to the MH. Meanwhile, if measured at lower layers (i.e. the phys-

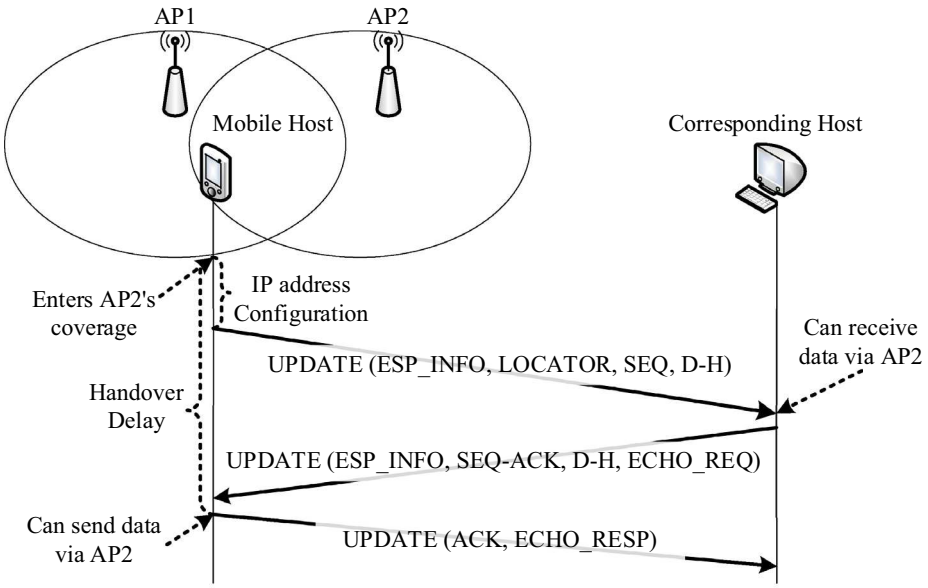


Fig. 2 HIP update sequence (three-way handshake). The CH can start receiving via AP2 after receiving the first UPDATE. The MH can start sending via AP2 after receiving the UPDATE from CH

ical layer), it is difficult to recognize the additional delay and packet loss due to the de-jitter buffer and the packet loss concealment (PLC) mechanism. Furthermore, by employing the transmission sequence number on the HIP header, the missing packets and the packet delays can be obtained. Additionally, the access point from where the packets originated can also be determined at the HIP layer. Thus, in the proposed method, HIP layer is opted for the MOS calculation to avoid the stated problems. Furthermore, the multi-homing and the UPDATE message are utilized to obtain the MOS values of the networks that the MH is currently connected to; this will be discussed in detail in the following subsections.

2.2 Modifications to the UPDATE Header for MOS Estimation

The handover approach in this study employs the same probing technique used in [15] where the MH sends a request for packet duplication to the CH. When the CH receives this request, the CH will send the duplicated VoIP packets to all paths available between the MH and CH. In [15], because they use SCTP, the request is added to their session update mechanism, where an ASCONF message is sent to the CH. Similar to that, in this work, the HIP header is modified by adding a single bit to indicate the request for starting or stopping duplication to the CH. From the duplicated messages, the one-way delay, jitter and packet loss rate can be estimated. This information are collected from a sample of 200 received packets from each AP during the packet duplication phase. Then using this information, the R-factor and the MOS value of the current and target networks can be estimated using the method discussed in Sect. 2.3.

2.3 E-Model Mean Opinion Score

E-Model is a model standardized as G.107 [25] by the International Telecommunication Union—Telecommunication Standardization Sector (ITU-T) to estimate the subjective quality of VoIP calls. The score is basically calculated using the Eq. (1), where R is known as the R-factor which signifies the quality level of the call; R_o is the signal to noise ratio; I_s indicates the impairments simultaneous to the voice signal transmission; I_d specifies the impairments caused by delay metrics; the I_e is the effects of equipment which is affected by the packet loss rate, and finally A is the advantage factor which is determined by the caller expectations.

$$R = R_o - I_s - I_d - I_e + A \tag{1}$$

$$R = 93.34 - I_d - I_e \tag{2}$$

The equation can be further simplified by using the default values for R_o, and I_s from [25] and A is assumed to be zero, meaning that the R is purely affected by the delay context and the packet loss rate. The simplified version of the R-factor equation is as shown by Eq. (2). The I_d can be estimated using the method introduced in [26]. The method uses the real-time protocol (RTP) one-way delay (obtained by subtracting the timestamp of the packet with the receiving time at MH) and the jitter which is obtained from Eqs. (3) to (5), and in turn, will be filtered using Eq. (6).

$$d_{\text{current}} \leftarrow t_{\text{current}} - t_{\text{timestamp}} \tag{3}$$

$$j_{\text{current}} \leftarrow d_{\text{current}} - d_{\text{previous}} \tag{4}$$

$$j \leftarrow j + \frac{1}{16}\left(j_{\text{current}} - j\right) \tag{5}$$

Table 1 The values for factor I_e under packet loss conditions for codecs G.711 without packet loss concealment (PLC) [28], G.711 with PLC [28], iLBC [37], SPEEX [37] and SILK [37]

Packet loss (%)	G.711	G.711 with PLC	iLBC	SPEEX	SILK
0	0	0	18.013	27.423	1.31953
1	25	5	23.13878	29.70327	8.773872
2	35	7	26.39093	31.62536	12.55644
3	45	10	28.77829	33.28666	15.11077
5	55	15	32.22617	36.05645	18.59509

t_{current}, current time; $t_{\text{timestamp}}$, the packet time stamp; d_{current}, current end-to-end delay; d_{previous} the previous end-to-end delay; j_{current} the current jitter. j, accumulated average jitter (inter arrival jitter).

d_{current} is the current end-to-end delay which is calculated by subtracting the current time with the time stamp. Then, the current instantaneous jitter, j_{current}, is obtained by subtracting d_{current} with the previous end-to-end delay, d_{previous}. Finally, the average jitter, or also known as the inter arrival jitter [27] is obtained from the moving average of the jitter as shown in Eq. (5). This algorithm [Eqs. (3)–(5)] was adopted from [27], where the value of 1/16 is the weight that determines the latest n observation of the instantaneous values of jitter (j_{current} of previous instances) that really influence the estimated inter arrival jitter. The number of jitter observations is ≈5 (1/weight coefficient); thus, in this case, the weight coefficient is $1/16 = 0.0625$ and the number of effective observations is 5 $(1/0.0625) = 80$. This means, the study in [27] recommends that using the latest 80 jitter observations are sufficient to effectively estimate the inter arrival jitter of a considered network.

$$D_{\text{j}} = \min\left(\text{codec_frame_size} + (0.9^{*}\text{RTP_jitter}), 300\right) \quad (6)$$

The Dj is actually dependent on the jitter buffer used in VoIP devices, and usually, a limit is configurable by the user; in Eq. (6), the limit is set to 300 ms.

Meanwhile, the value of I_e is determined according to Table 1, which is obtained from ITU-T standard G.113 [28]. The packet loss is detected by checking the number of missed transmission sequence number TSN, and the percentage of packet loss is calculated by dividing the accumulated lost packet per the number of the current TSN. Then, the value is compared to Table 1 to determine the value of I_e of the network.

Then, the R-factor is converted to MOS using Eq. (7) to rate the MOS level of each path between the MH and the CH [25]. The level of satisfaction by the user is listed as in Table 2, where the score of 4.34 and above indicates the best call quality experienced, while the value of below 3.1 means that the quality experienced is very poor [25]. This MOS value will be used in the proposed method as one of the context when deciding the handover trigger timing.

Table 2 The mean opinion score experience classification

R value (lower limit)	MOS (lower limit)	User perception
90	4.34	Very satisfied
80	4.03	Satisfied
70	3.6	Some users dissatisfied
60	3.1	Many users dissatisfied
50	2.58	Nearly all users dissatisfied

$$\text{MOS} = \begin{cases} 1 & R < 0 \\ 1 + 0.035R + R\,(R-60) & \\ \quad (100 - R)\left(7 \times 10^{-6}\right) & 0 < R < 100 \\ 4.5 & R > 100 \end{cases} \quad (7)$$

2.4 The Ant Colony Probabilistic Equation

The Ant System (AS) [29] and the Ant Colony Optimization (ACO) [30], which fall under the category of metaheuristic biologically-inspired Swarm Intelligence [31,32], were inspired by the foraging (the act of searching for food or provision) behavior of some ant species. While searching for food, an ant will deploy a chemical substance known as pheromone on the ground that it has roamed on. When it reaches the destination (in this case food source), it will retrace back on the same route where it has taken (the one with the pheromone) to return back to its nest. Then, other ants will follow the same path toward the food source. A summary of this process is as shown in Fig. 3. Let the upper path be shorter than the lower path. As time goes by, the density of pheromone on the upper path will become apparently higher than the lower path, influencing other ants to choose the upper path due to the pheromone intensity.

This concept is useful in determining the best path, however, to implement this concept into a wireless network scenario, some additional information besides the pheromone level is considered. Some implementations of this concept have been proposed in the wireless sensor network (WSN) research and have shown exceptional results [33–35]. This is part of the motivation to implement such methods

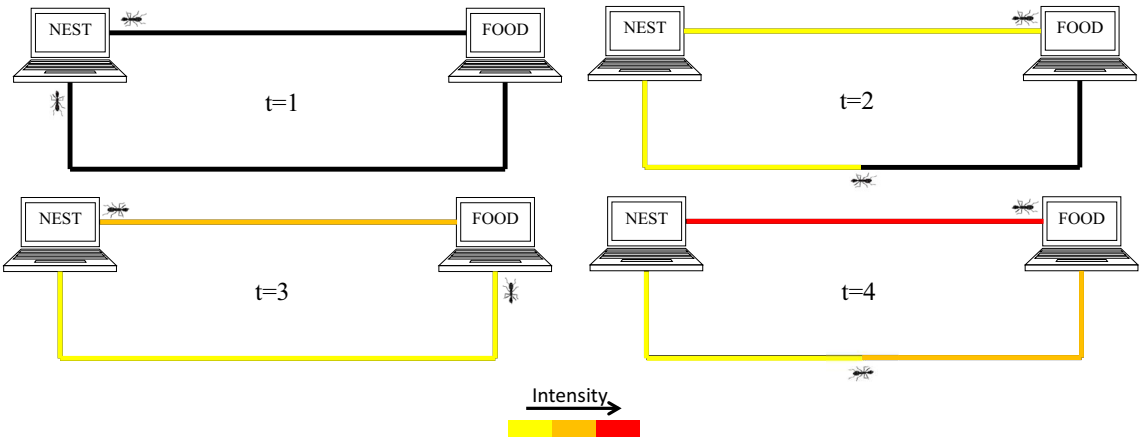


Fig. 3 Ant Colony basic concept of pheromone deployment; the intensity of pheromone increases with the frequency that the ant uses the path

in this work. The implementation of the Ant Colony (AC) concept is done using the probabilistic equation shown by Eq. (8).

$$p_i\,(t) = \frac{[\eta_i\,(t)]^{\alpha}\,.\,[\tau_i\,(t)]^{\beta}}{\sum_{i=1}^{N}[\eta_i\,(t)]^{\alpha}\,.\,[\tau_i\,(t)]^{\beta}} \tag{8}$$

$p_i\,(t)$ is the overall desirability of MH to choose path i. $\tau_i\,(t)$ is the pheromone level which is the normalized MOS level in the proposed handover approach. $\eta_i\,(t)$ is a heuristic evaluation of path i; in this study, it represents the normalized RSS value. α and β are the relative weights for $\eta_i\,(t)$ and $\tau_i\,(t)$ respectively. The values of α and β ranges from 0 to 1 and the sum of them equals to 1 ($\alpha + \beta = 1$). How to utilize this equation will be discussed in the following section. In this form, there are only two parameters considered, but depending on the number of contexts that a method uses, the number of heuristic components can be added accordingly.

3 The Proposed Handover Approach

The proposed method in this manuscript is closely related to the ECHO [15] because the probing mechanism used for the two methods are similar. In addition, the architectures of ECHO and the proposed approach are similar to SIGMA [12]. The main difference is that the proposed method uses HIP in the HEP and also the implementation of ACPE in the HDP. Figure 4a shows the flowchart of ECHO, while Fig. 4b shows the flowchart of the proposed method.

In brief, ECHO starts checking the MOS level of both, the target as well as the current serving, networks when the RSS level of the current network becomes lower than the alternative network. Then, the MH checks whether the MOS of the alternative network is higher than the MOS threshold (a threshold value of 3.1 was selected in [15] which is the second lowest item in Table 2). If this condition is met, the handover is initiated. If not, no handover occurs and the MH will continue to monitor the RSS of the current main network, and if it falls below the RSS threshold (in [15] it is set to −79 dBm) a handover is triggered, even though the target network does not support VoIP call. Some disadvantage in this flowchart is that the handover will only occur after the RSS of the current network falls below the target network, and the handover is triggered according to the threshold regardless of the situation when the current network still has better MOS level compared to the target network. This is partly due to the late probing and the threshold as well as the fact that the MOS and RSS are considered separately.

To alleviate this issue, the proposed approach starts the probing when the MH completes its association with the new network, then combines the RSS information and MOS value using the ACPE. The flowchart of the proposed method is as shown in Fig. 4b. The MH monitors the ACPE value (P_main, P_target) and RSS of the current serving cell as well as the target cell. P_main is the value of $p_i\,(t)$ of the main network, while P_target is the value of $p_i\,(t)$ of the target network. If the value of P_target is higher than P_main + hysteresis (the value used for hysteresis in this work is 0.01), the handover process to the target cell will be initiated. However, if P_target is lower than P_main + hysteresis, the MH will continue to use the current network, and no handover is triggered. In the meantime, MH continues to monitor P_main and P_target. Then, if P_main + hysteresis falls below P_target or the RSS of the current serving network falls below the RSS threshold (in the simulation, −81 dBm was used as the RSS threshold), handover to target network will be initiated, regardless of the values of P_main and P_target.

The flowchart in Fig. 4b shows the best case scenario where the handover message is assumed to always be successfully received by the CH. However, in the real world, this is not the case. To cater for the scenario where the handover message is lost during the handover process, handover

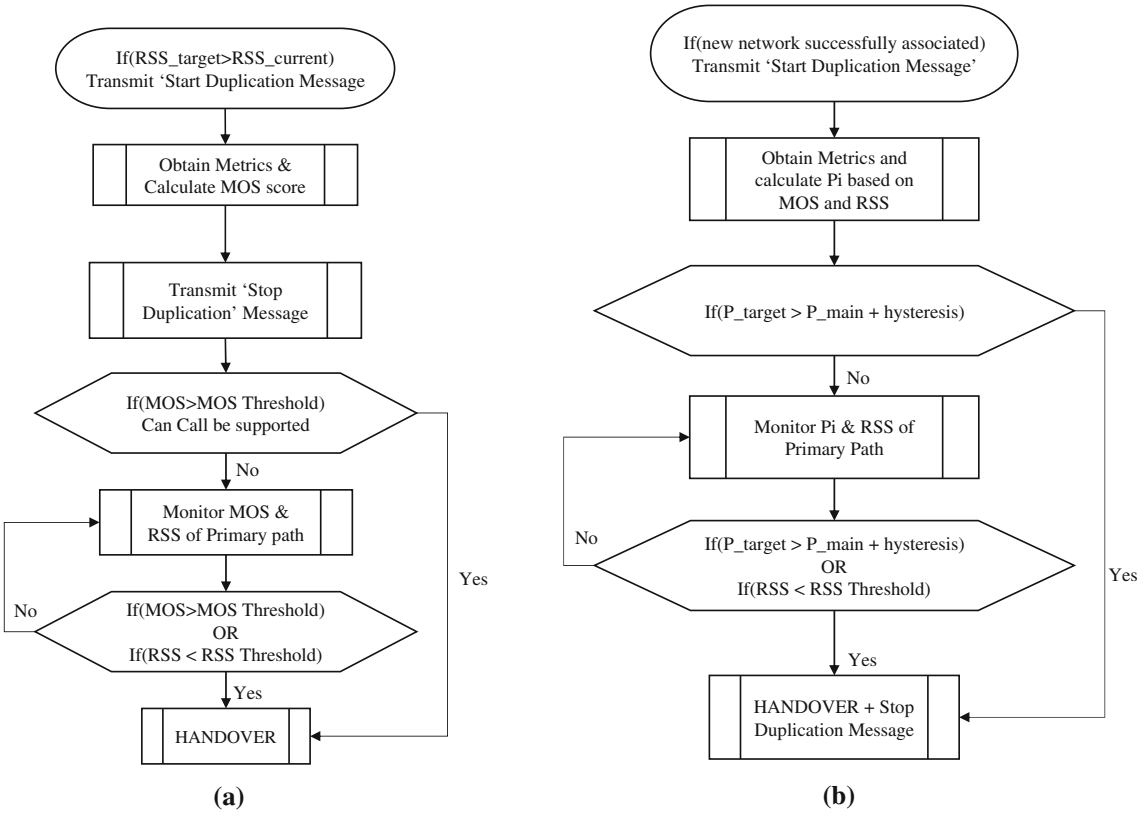


Fig. 4 **a** The ECHO flowchart, **b** the proposed method's flowchart

message retransmission timeout (RTO) is implemented in the recovery phase. When a handover message is sent, a timeout counter is initiated, and if the handover message is not acknowledged within the interval before the timeout expires, the handover message will be re-sent to the CH. Since one-way delay (OWD) information is collected for the MOS calculation, the same value can be used to calculate the retransmission timeout (RTO) of the handover message similar to the RTO calculation method used for SCTP. Assuming that the data path is symmetric, the round trip time (RTT) is estimated as 2OWD. Then, the RTO calculation similar to SCTP can be calculated using the following equations.

For first time measurement:

$$\mathrm{SRTT} = \mathrm{RTT}_{1\mathrm{st}} \tag{9}$$

$$\mathrm{RTTVAR} = \mathrm{RTT}_{1\mathrm{st}}/2 \tag{10}$$

$$\mathrm{RTO} = \mathrm{SRTT} + 4\mathrm{RTTVAR} \tag{11}$$

For first time measurement, the smoothed RTT is equals to the first RTT. The RTTVAR is equals to $\mathrm{RTT}_{1\mathrm{st}}$ divided by 2. The RTO is the sum of SRTT and 4RTTVAR. Then, the consecutive measurements of RTO are as follows:

$$\mathrm{RTTVAR}_{\mathrm{new}} \leftarrow (1-\gamma)\,\mathrm{RTTVAR}_{\mathrm{old}} \\ \gamma\,(\mathrm{SRTT}_{\mathrm{old}} - \mathrm{RTT}_{\mathrm{new}}) \tag{12}$$

$$\mathrm{SRTT}_{\mathrm{new}} \leftarrow (1-\omega)\,\mathrm{SRTT}_{\mathrm{old}} + \omega\mathrm{RTT}_{\mathrm{new}} \tag{13}$$

$$\mathrm{RTO} = \mathrm{SRTT}_{\mathrm{new}} + 4\mathrm{RTTVAR}_{\mathrm{new}} \tag{14}$$

γ and ω are constants and the recommended values are 0.25 and 0.125 respectively. Using this mechanism, the proposed handover approach will be able to recover if the handover message is lost during the handover process. In the next section, the simulation settings and the results will be discussed thoroughly.

4 Results and Discussions

The proposed method and ECHO are simulated using OMNeT++ version 4.1 with the HIPSim++ module [36]. The network topology considered in this study is as shown in Fig. 5, where the MH is programmed to move along the path shown by the dotted arrow. The network parameters used for the simulation are as listed in Table 3. In the simulation, IEEE802.11b and 11g are used due to the RAT availability of the simulator. However, we assume that the proposed work is applicable to any different RATs because the implementation is on the network layer and above; hence, it is not affected by the difference on the MAC and PHY layer. Since we are considering VoIP services, the network traffic used in

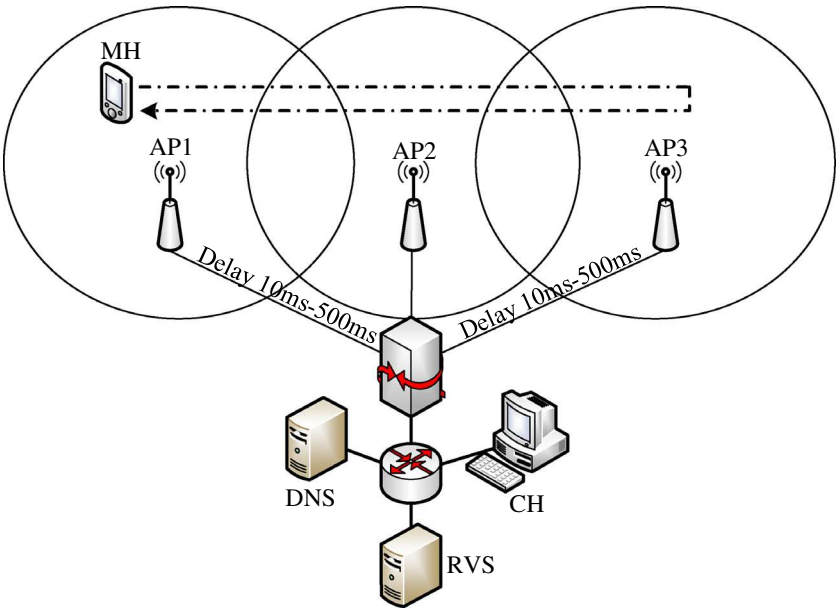


Fig. 5 Simulation topology for the simulation

Table 3 Significant configuration of the access network

AP coverage diameter	170 m
Distance between two APs	100 m
Overlapping distance	140 m
Link delay at AP2	10, 450 ms
Link delay at AP1 and AP3	10, 50, 100, 150, 200, 250, 300, 350, 400, 450, 500 ms

the simulation is in accordance with the G.711 codec, where 160 bytes packets are sent every 0.02 s which amounts to 64 kbps. The MH moves at a speed of 5 m per second for all the simulation. The discussion on the results will be divided into three parts. For the first discussion, the link delay on AP2 is set to 10 ms and the delays on AP1 and AP3 are varied from 10 to 500 ms. For the second discussion, the delay for AP2 is configured to 450 ms and the delays for AP1 and AP3 are also varied from 10 to 500 ms. The last part will discuss the behavior of ECHO and the proposed method and the reason why there are difference in the performance. As a reminder, α is the weight for RSS and β is the weight for MOS in the ACPE.

4.1 AP2 Delay = 10 ms, AP1 and AP3 Delay = 10–500 ms

Figure 6 shows the average MOS simulation results for the proposed approach and ECHO. The vertical axis is the average MOS for the whole simulation run. Take note that in this section and in Sect. 4.2, the vertical axis of the graphs is the average MOS for the whole simulation run, and not the instantaneous MOS used for the handover triggering. The horizontal axis is the delay settings on AP1 and AP3 (namely AP1/AP3). The delay on AP2 is set at a static value of 10 and 450 ms for the first and the second scenarios, respectively. In Fig. 6, the simulation results of the proposed method and ECHO are shown as plots 'plus' and 'diamond', respectively, when the delay on AP2 $=$ 10 ms. Meanwhile, when the delay at AP2 $=$ 450 ms, the result of the proposed method and ECHO are shown as plots of 'cross' and 'circle', respectively. A 95 % confidence interval is included in the graph. However, since the value is very small, it is not very visible.

In the first scenario, from the graph, it can be seen that generally, when the delay increases, the average MOS for each configuration drops. The proposed method has better or nearly the same performance as ECHO for all weight configuration up till 200 ms of delay. After that, the proposed method is superior to ECHO in some weight configuration. Out of all the weight configurations, the combination of $\alpha = 0.1$ and $\beta = 0.9$ has shown the best outcome. From the same plots, it can be seen that when the disparity of the link delay is small, the average MOS of both methods are nearly the same. However, when the difference of delay between AP2 and AP1/AP3 becomes larger, the performance difference becomes clear. The proposed method can maintain a higher overall average MOS even when handing over to a network with low MOS levels.

4.2 AP2 Delay = 450 ms, AP1 and AP3 Delay = 10–500 ms

In the second scenario, the link delay on AP2 is set to 450 ms while the link delays on AP1 and AP3 are varied from 10 to 500 ms. This time, the performance of the proposed method becomes nearly the same as ECHO when the delay on AP1/AP3 increases because the MOS disparity between the networks is decreasing as shown in Fig. 6. From this

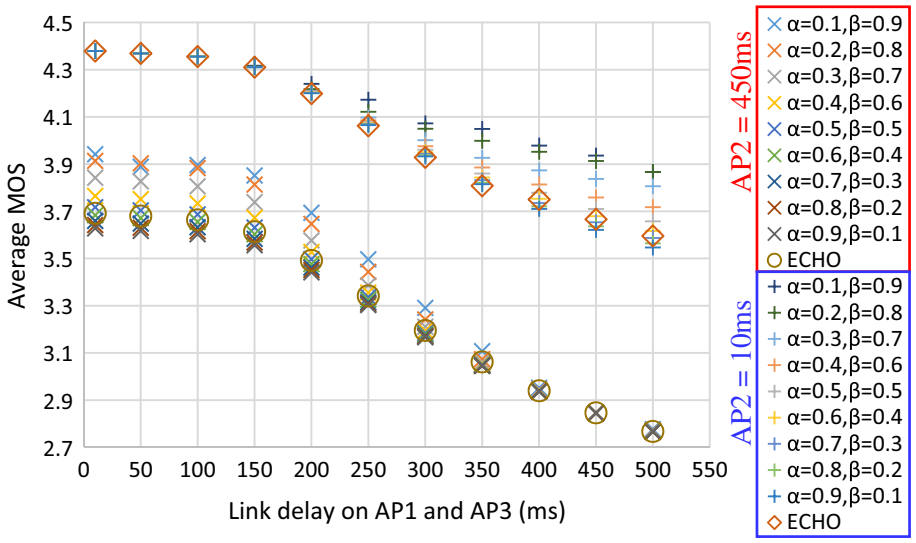


Fig. 6 Average MOS comparison between the proposed method and ECHO for different weight settings. The graph with *plus plots* representing the proposed method and *diamond plot* representing ECHO are showing the simulation results when the delay on AP2 = 10 ms. Meanwhile, the graph with *cross plots* representing the proposed method and *circle plot* representing ECHO are showing the simulation results when the delay on AP2 = 450 ms. The delays on AP1 and AP3 are varied from 10 to 500 ms for all simulation

graph, the proposed method with weight configurations of $\alpha = 0.1$ and $\beta = 0.9$ is still the best configuration. Here, we can see that the proposed method maintains a higher average MOS compared to ECHO when the difference in terms of link delay between AP2 and AP1/AP3 is large. One of the significant points is when the delay on AP1/AP3 is set to 200 ms, the average MOS level of the proposed method is one level higher than ECHO when referred to Table 2. However, when the delay on AP1/AP3 is large, the average MOS for both approaches are similar. This is inevitable, since all networks have very large delay, but notice that the proposed method is still slightly better when the delay of AP1/AP3 drops to 500 ms.

4.3 Handover Triggering Timing Comparison

In this subsection, the outcomes of the previous two scenarios stated in the previous subsection will be discussed. Figure 7 shows how the instantaneous MOS changes using ECHO and the proposed method in the first scenario during a portion of the. The horizontal axis is the simulation time in seconds, and the vertical axis is the instantaneous MOS used for the handover trigger. Each colored plot (e.g., 10, 5, …, 500 ms) represents the delay on AP1/AP3. The main point to focus on this result is the path switch timing pattern of both approaches. The drop in MOS in this graph is due to handing over to the alternative network with higher delay (e.g., AP2 to AP3). Notice that as the delay on AP1/AP3 increases, the duration the MH stays within the network with lower MOS becomes shorter when using the proposed method. For example, when the delay on AP1/AP3 is 150 ms correspondingly, the duration that the MH connects to the network with lower MOS is around 40 s compared to just 20 s when the delay on AP1/AP3 are 500 ms correspondingly.

When compared with ECHO, it can be seen that the proposed approach triggers the handover to the alternative network with better MOS earlier when the MH is moving

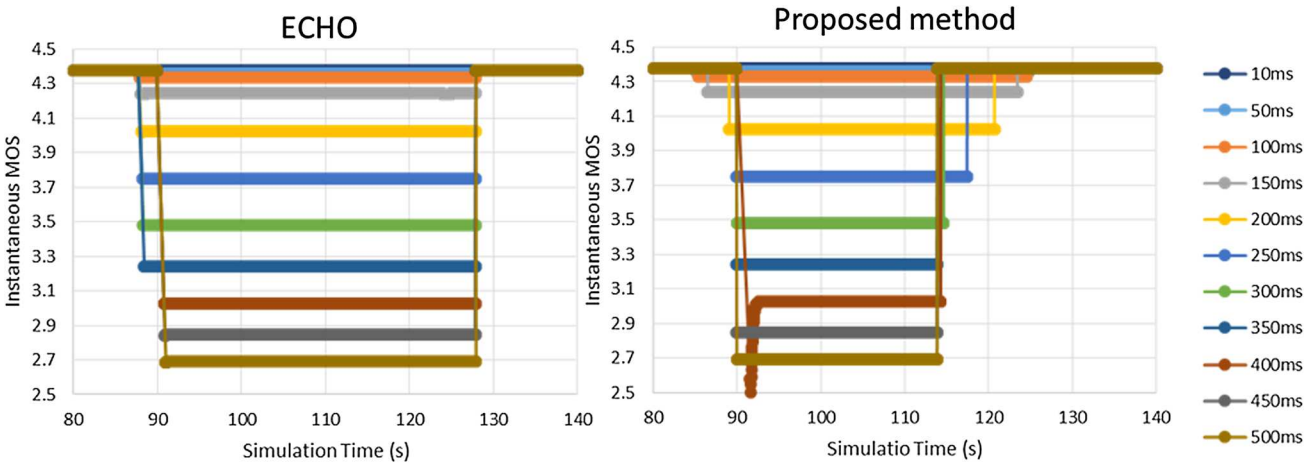


Fig. 7 Handover timing comparison between the proposed method and ECHO

from the network with low MOS to a network with higher MOS. For example, when the delay on AP1/AP3 is 300 ms, the proposed approach triggers the handover at around the 115 s instance of the simulation. On the other hand, ECHO triggers the handover at 128 s instance of the simulation regardless of the MOS of the current network. The inverse effect can be seen when the MH is moving from the network with high MOS to the network with lower MOS.

In summary, it can be seen that the proposed approach reduces its connection time with the network with low MOS and increases its connection time with the network with higher MOS. Meanwhile, ECHO is restricted to the static threshold settings reducing the efficiency in maintaining a higher overall average MOS.

4.4 The Decision Tendency of the Proposed Method

The proposed method uses the ACPE to combine the MOS value with the value of RSS during the HDP. The values of MOS and RSS are normalized and the weighting coefficient of each parameter, α and β, also ranges from 0 to 1. For example, let $\eta = 0.2$, $\tau = 0.1$, $\alpha = 0.1$ and $\beta = 0.9$. Mathematically $\eta^{\alpha} = 0.8513$ and $\tau^{\beta} = 0.126$. From these values, one might think that due to the larger value of η^{α} compared to τ^{β}, the tendency of the ACPE equation, Eq. (8), will follow the tendency of η. However, this is not true. Take note that $p_i(t)$ is a product of η^{α} and τ^{β}, and these parameters are governed by the laws of indices; any parameter with an index of 1 is equivalent to the number itself, while any parameter with an index of 0 is equivalent to 1. Extending this concept, any parameter with an index closer to 1 will have a value closer to the parameter's original value, while any parameter with an index closer to 0 will have a value closer to 1.

In this manuscript, the important point in deciding the handover is not the parameter itself, but the combination of the considered parameters, which is the product of both parameters η^{α} and τ^{β}. Taking the best α and β configuration into consideration, when $\alpha = 0.1$, it means that the parameter with the index of α will have a value closest to 1, while $\beta = 0.9$, it means that the parameter with the index of β will have a value closest to its original value. Taking this fact into consideration, any number multiplied by 1 equals to its original value; any number multiplied by a number close to 1, equals to a value close to its original value. Thus, in this case, the value of $p_i(t)$, which is a resultant value from the product of τ^{α} and η^{β}, will have a value closer to the original value of τ, the MOS parameter. Hence, from this deduction, it can be concluded that the parameter with higher value of index will influence the tendency of $p_i(t)$ in this case.

This can be proven by observing the tendency of η, τ and $p_i(t)$ from Fig. 8. The first graph (top left) depicts the situation where only η is considered ($\alpha = 1$ and $\beta = 0$). The second graph (top right) shows the tendency of $p_i(t)$ when only τ is considered ($\alpha = 0$ and $\beta = 1$). The third graph (bottom left) illustrates the $p_i(t)$ with configurations of ($\alpha = 0.1$ and $\beta = 0.9$) and finally ($\alpha = 0.9$ and $\beta = 0.1$) are shown in the last graph (bottom right). The vertical axis shows the instantaneous $p_i(t)$ and the horizontal axis shows the simulation time in seconds. These graphs show the values of $p_i(t)$ for the main NIC and the secondary NIC. In the simulation, 2 NICs, NIC1 and NIC2, are used. Either of them can become the Main or Secondary NIC. Notice that the values of the Main NIC is always on the upper half (more than 0.5) because, when the target network becomes the Main Network, the previous 'Secondary' NIC will become the current 'Main' NIC. The graph legend in Fig. 8 shows the representation of each line color. The legend with 'pMain' shows the representation of the main NIC value, while 'pSec' shows the representation of the secondary NIC instantaneous value of $p_i(t)$ for each delay configuration on AP1/AP3. For example, 'ajse-pSecVecS0 = 10 ms' shows the plots for secondary NIC when the delay on AP1/AP3 is 10 ms. Furthermore, in this simulation, the link delay on AP2 is configured to 10 ms, while the delay on AP1/AP3 is varied from 10 to 500 ms.

As a reminder, in the simulation, the varied parameters are the delay on AP1/AP3, which in turn affects the MOS of each network. Thus, when the MH only considers RSS for the handover trigger, the triggering time (when the lines intersects at 0.5) for different scenarios is similar to each other since RSS is not affected by different network delays. In the first graph, it can be seen that the handover trigger time (the $p_i(t)$ crossings between pMain and pSec) is similar for all delay configurations.

In the second graph, it can be seen that the tendency of $p_i(t)$ is different for each simulation run, since only the MOS parameter, which is affected by the changes in the delay of each network, is considered. From this figure, it is clear that when MH considers MOS as a decision parameter, the delay on a network influences the value of $p_i(t)$. Up till this point, the graphs shows the tendency of $p_i(t)$ when MH only considers one parameter.

The last graph shows the value of $p_i(t)$ when the configuration is $\alpha = 0.1$ and $\beta = 0.9$. From this graph, it can be seen that the tendency of $p_i(t)$ is affected by the change of MOS as shown in the second graph, and at the same time shows some semblance to the first graph. As deducted previously, using this configuration, the value of $p_i(t)$ will be close to the value and changes of MOS, since the value of τ^{α} is near to 1 due to the index value of $\alpha = 0.1$. From the graph, it can also be seen that the handover trigger time changes due to the changes of MOS as well as RSS. This point is important since these differences in triggering time is what determines the total outcome of the proposed method as discussed in the previous section. When the MH is connected to a network with low MOS, it will handover to a network with higher MOS as soon as possible. Conversely, when a MH is con-

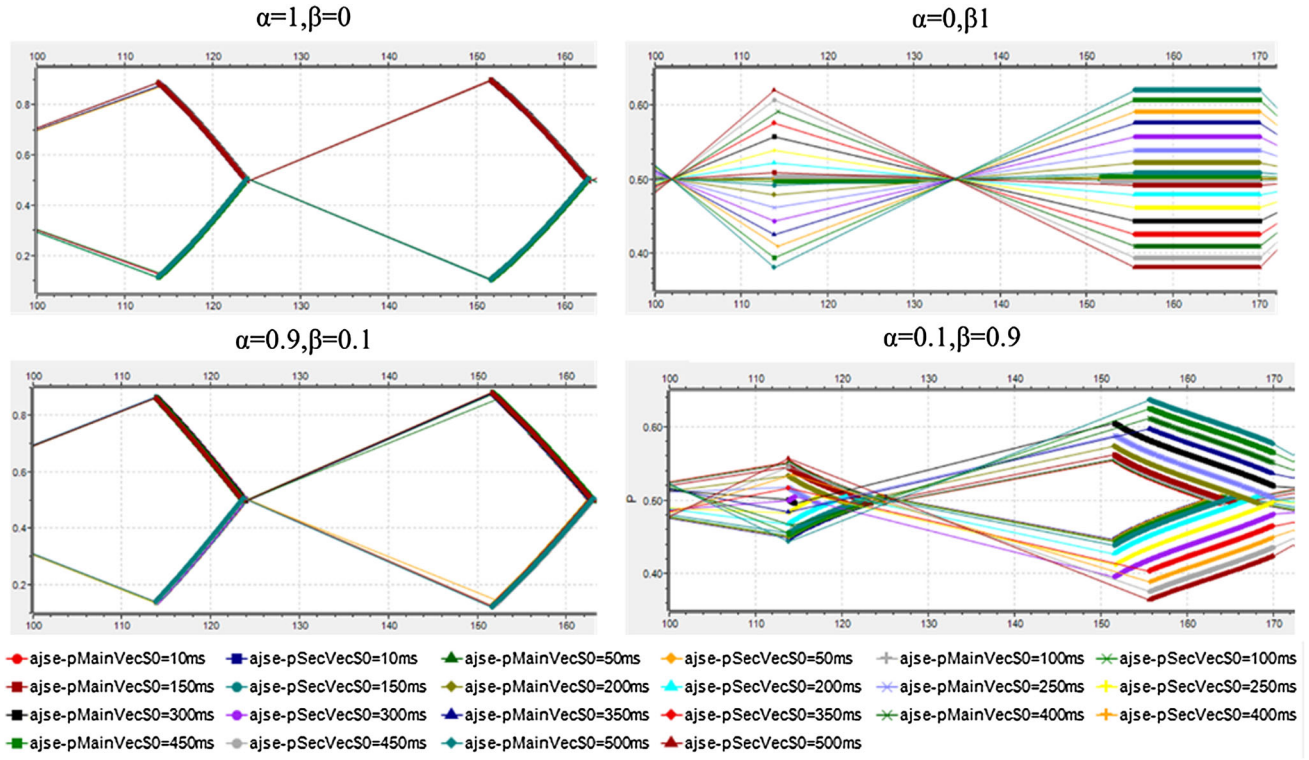


Fig. 8 The decision tendency comparison of the proposed method. The first graph (*top left*): $\alpha = 1, \beta = 0$; second graph (*top right*): $\alpha = 0, \beta = 1$; third graph (*bottom left*): $\alpha = 0.9, \beta = 0.1$; last fourth graph (*bottom right*): $\alpha = 0.1, \beta = 0.9$. The *vertical axis* is the $p_i(t)$, while the *horizontal axis* is the simulation time (s)

nected to a network with high MOS, it will handover to a target network with lower MOS as late as possible.

Finally, the third graph shows the value of $p_i(t)$ when the configuration of $\alpha = 0.9$ and $\beta = 0.1$ is used. From this graph, it can be seen that the outcome is similar to the results in the first graph. Even though in this case, the value of τ^{β} is larger, the value is actually nearing to the value of 1, while the value of η^{α} is nearing the original value of η; thus, the graph becomes similar to that of the first graph.

In conclusion, when the index of a parameter is nearing to the value of 1, and the value of the parameter is normalized; then, the resulting value will be close to that parameter's original value. Meanwhile, when the index of a parameter is nearing to the value of 0, and the value of the parameter is normalized, the resulting value will be close to 1. Finally, when these two values are multiplied together, the resulting value will be close to the parameter with the higher index value.

4.5 The Effects of Packet Loss and Different Speech codecs

In this section, the effects of packet loss rate and the effects of different speech codecs on the performance of the proposed method is discussed. The analysis is done for packet loss rates of 0 % (base configuration), 1, 2, 3, and 5 %. The packet loss rate is limited to just 5 % because the recommended value of I_e used for factor-R and MOS calculation for G.711 codec without PLC is provided only up till 5 % packet loss. In MOS calculation, the values of I_d (impairments due to delay) are affected by the end-to-end delay and jitter; I_e (effective equipment impairment factor), on the other hand, is affected by the packet loss rate and the type of codec used in the VoIP call. The values of I_e for G.711 without PLC and for G.711 with PLC are already provided by ITU-T in G.113. However, the I_e values of other codecs are not readily available. In this work, the values of I_e for non ITU-T codecs were obtained from the fitting equation derived by Assem et al. [37]. From their study, the I_e values of iLBC, SPEEX and SILK can be obtain using the following equation:

$$I_e = a \log\left(1 + b \times \text{packetLoss}_{\%}\right) + c \tag{1}$$

where the values of a, b and c are as shown in Table 4. Using this equation, the values of I_e for iLBC, SPEEX and SILK can be obtained, and these values are used in the simulation to compare the performance of the proposed method when equipped with these codecs. The I_e values of the considered speech codec are shown in Table 1.

Table 4 The values of a, b and c for the corresponding codecs

Parameters	iLBC	SPEEX	SILK
a	20.836	28.244	18.3442
b	0.762	0.2043	1.5489
c	18.013	27.423	1.3195

The first analysis is to study the effects of packet loss rate on the total average MOS performance of the proposed method and ECHO. This analysis is done using the G.711 codec without PLC for all configurations. The second analysis is to study the effects of using different codecs to the total average MOS performance. This analysis varies the codec used in order to compare the performance with different codecs.

The results of the first analysis are shown in the following Fig. 9. This figure shows the results when $\alpha = 0.1$, $\beta = 0.9$, using G.711 codec without PLC, **link** delay on AP2 = 450ms, and the percentage of packet loss rate is varied (0, 1, 2, 3 and 5 %). The packet loss only occurs on AP2's network. In general, the total average MOS becomes lower as the of packet loss rate increases. In Fig. 9, it can be seen that when there are no packet drops, the proposed method is better than ECHO only when the link delay of AP1/AP3 is below 400 ms. However, as the packet loss rate increases it can be seen that the proposed method outperforms ECHO considerably for all AP1/AP3 delay configurations. In a nutshell, the proposed method can outperform ECHO when the packet loss rate at AP2 increases.

In the second analysis, the performance of different types of codecs (ITU-T: G.711 without PLC and G.711 with PLC; non ITU-T: iLBC, SPEEX, SILK) are compared when the packet loss rate at AP2 changes and the link delay of AP1/AP3 is varied from 10 to 500 ms. The link delay on AP2 is configured to be 10 ms for all simulations. The results are shown in Fig. 10. The data rate used for each codec is based on its maximum (or nearly maximum) supported data rates as listed in Table 5. The graph in Fig. 10 shows the average plots of each codec for all packet loss rate configuration. The 95 % confidence intervals are very small and negligible. The error bar in the graph shows the range of values that each codec can arrive at with different packet loss rates. For example, when the packet loss rate is 0 %, the value achieved by each codec is at maximum. Meanwhile, when the packet loss rate is 5 %, the value is at minimum for each codec. In a nutshell, the error bar is inversely proportionate to the packet loss rate: the higher the packet loss rate, the lower the average MOS value obtained by each codec.

From the graph in Fig. 10, it can be generally seen that the total average MOS for all codec configurations drops as the packet loss rate increases. When there are no packet loss it can be seen that G.711 codec with PLC, G.711 codec without PLC and SILK have similar performance, followed by iLBC and SPEEX has the worst performance. However, as the percentage of packet loss at AP2 increases, the performance of G.711 codec without PLC drops drastically; where at 5 % packet loss rate it is the worst compared to other codecs. The performance of other codecs, meanwhile, only decreases slightly with each increment of packet drops. From here, it can be clearly seen that different codec will yield different performance levels and the value of I_e obtained from different codecs greatly influences the total performance of a VoIP call. Furthermore, it can also be concluded that the G.711 codec with PLC, iLBC and SILK are suitable for implementation in the real VoIP application, because the performance can still be maintained above 3 point of the MOS scale (which means the experience is still acceptable), even though the packet loss rate is as high as 5 % and the end-to-end delay is high (500 ms, which is the maximum acceptable one-way delay

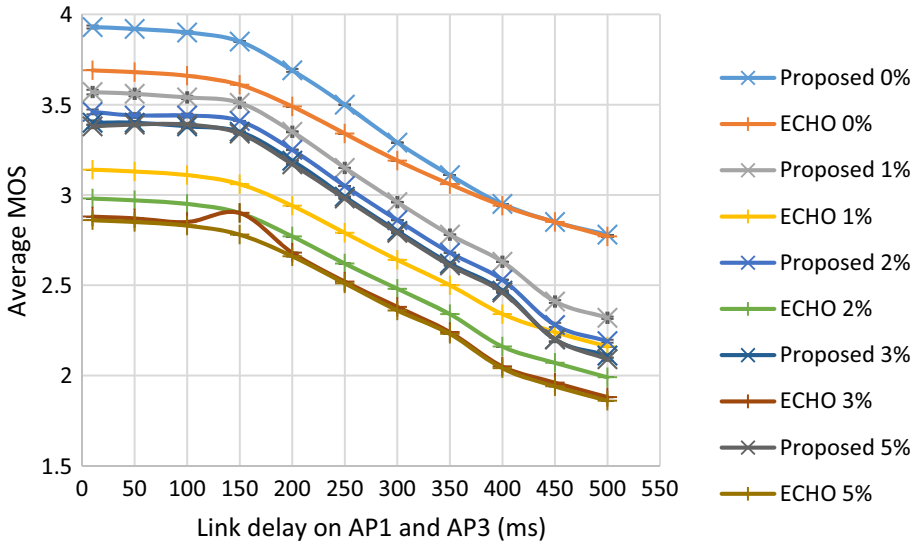


Fig. 9 The average MOS comparison of the proposed method and ECHO. The general configuration for this simulation is as follows: $\alpha = 0.1$, $\beta = 0.9$, G.711 without PLC, $AP2 = 450$ ms. The *cross plots* are the proposed method's plot with corresponding packet loss percentage. Meanwhile, the *plus plots* are the plot for ECHO with corresponding packet loss percentage

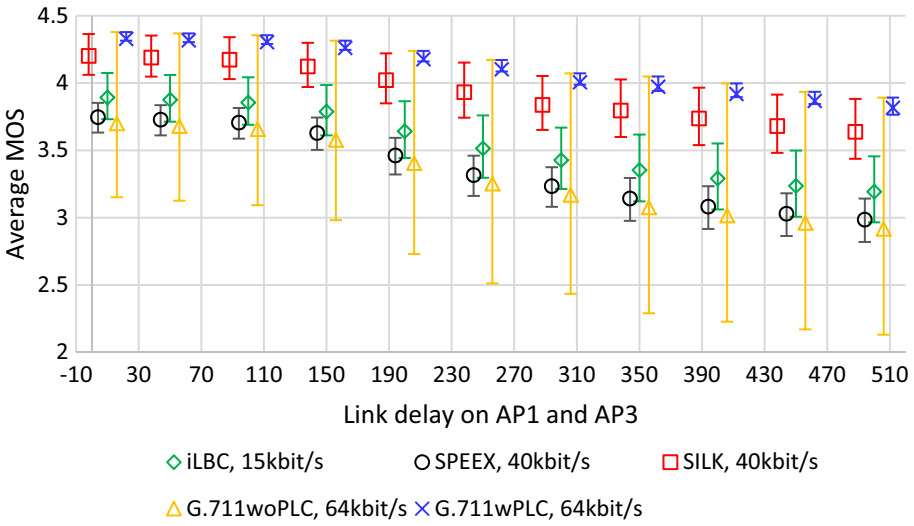


Fig. 10 The average MOS performance comparison of different codecs (G.711 codec with PLC, G.711 codec without PLC, iLBC, SPEEX and SILK). The general configurations for this simulation are: $\alpha = 0.1$, $\beta = 0.9$, AP2 = 10 ms. The *plots* show the averaged average MOS with different packet loss rate configuration for each codec considered. The *error bar* is at the peak when the packet loss rate is configured as 0 %, while the error bar is at the bottom when the packet loss rate is configured as 5 %

Table 5 The data rate configuration for each codec

Codec	Data rate used (kbps)
G.711 without PLC	64
G.711 with PLC	64
iLBC	40
SPEEX	40
SILK	15

for a VoIP call). From these analyses, it can be concluded that choosing an appropriate codec influences the quality of experience of the users.

5 Conclusion

In this paper, a new handover technique, which uses the Ant Colony Probabilistic Equation to combine the received signal strength and the MOS of the networks available to the Mobile Host for triggering a handover, has been proposed. A simulation on OMNeT++ was conducted using the proposed approach and an existing method known as the Endpoint Centric Handover (ECHO). The results showed that the proposed method can achieve better average MOS performance compared to ECHO when the disparity of link delay between the current serving network and the target network is large. Furthermore, it can be concluded that the proposed approach increases its connection time to a network with high MOS level and decreases its connection time with a network with low MOS level, compared to the corresponding connection time in ECHO, thus showing better overall average MOS than ECHO.

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